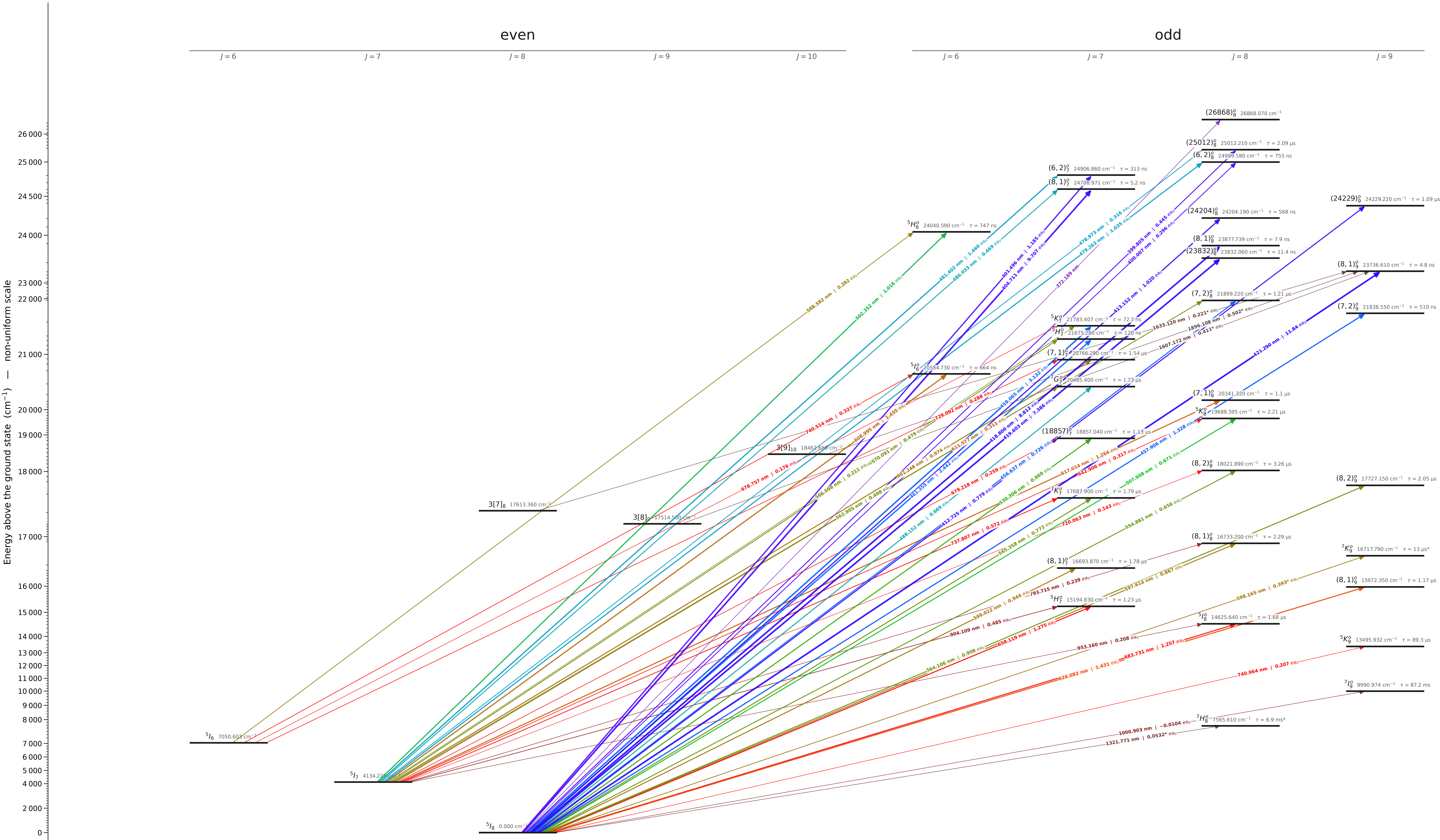
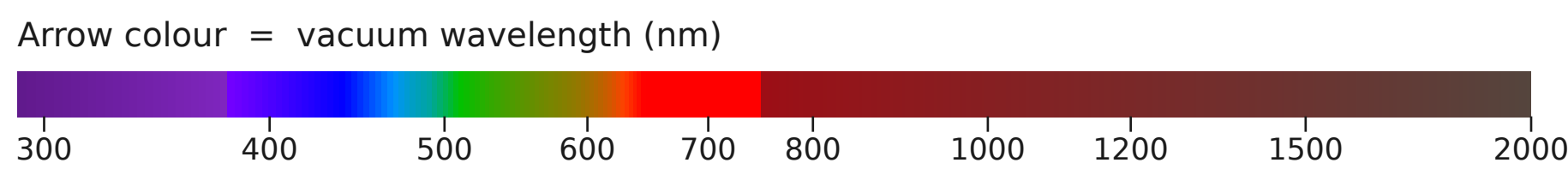


¹⁶³Dy (Dy I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 39 levels, 55 lines.
Dy⁺ ionisation limit 47901.76 cm^{-1} = 5.93906 eV (208.761 nm)

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Arrow label: vacuum wavelength reduced dipole matrix element $\{ \langle \psi || er || \psi' \rangle \}$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U\rangle$ (eao)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^1S_0 - (26868)^3S_0$	372.1890	372.0832	805.4845	-	-	-	-	UV ground-state line
$^1S_0 - (8, 1)^1J_0$	404.7113	404.5970	740.7563	9.707	1.920e+08	30.61	99.8	strong blue $J=8 \rightarrow 7$ line
$^1S_0 - (8, 1)^1J_0$	418.8001	418.6821	715.8366	8.812	1.260e+08	20.15	99.5	strong blue $J=8 \rightarrow 8$ line
$^1S_0 - (23832)^3J_0$	419.6028	419.4846	714.4672	7.386	8.800e+07	13.96	100	strong blue $J=8 \rightarrow 8$ line
$^1S_0 - (8, 1)^1J_0$	421.2902	421.1715	711.6057	11.84	2.000e+08	33.16	100	main broad cooling line: Zeeman slower, transverse coolin...
$^1S_0 - (7, 2)^1J_0$	456.6373	456.5093	656.5221	0.726	6.600e+05	0.1315	79.9	pure 6s $\rightarrow$ 6p reference line for King plots
$^1S_0 - (8, 2)^1J_0$	564.1065	563.9499	531.4466	0.908	4.900e+05	0.07764	100	closed $J=8 \rightarrow 9$ narrow line
$^1S_0 - ^7K^0_0$	598.1652	597.9985	501.1867	0.393*	7.700e+04	0.01224	100	"598 nm" closed $J=8 \rightarrow 9$ narrow line proposed for narrow-li...
$^1S_0 - (8, 1)^1J_0$	599.0223	598.8564	500.4696	0.944	5.600e+05	0.08941	99.7	narrow $J=8 \rightarrow 7$ probe line
$^1S_0 - (8, 1)^1J_0$	626.0819	625.9088	478.8390	1.431	8.900e+05	0.136	100	narrow-line MOT
$^1S_0 - ^3H^0_0$	658.1186	657.9368	455.5295	1.275	7.700e+05	0.1294	94.7	narrow probe line used for Doppler thermometry of the 421...
$^1S_0 - ^5J^0_0$	683.7308	683.5421	438.4657	1.257	5.893e+05	0.09474	99	narrow $J=8 \rightarrow 8$ optical pumping / demagnetization cooling l...
$^1S_0 - ^5K^0_0$	740.9640	740.7600	404.5979	0.207	1.120e+04	0.001782	100	ultra-narrow-line MOT / narrow-line cooling in tweezers
$^1S_0 - ^1J^0_0$	1000.9034	1000.6291	299.5219	0.010	1.150e+01	1.825e-06	100	ultra-narrow
$^1S_0 - ^1H^0_0$	1321.7705	1321.4091	226.8113	0.053*	1.460e+02	2.307e-05	100	proposed ultra-narrow telecom-band line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-163	-44(10)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-162	-178(13)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-161	-194(13)
$^1S_0 - ^3P_0$ (2686.8) μ m	372.189	164-160	-339(13)
$^1S_0 - (^8, ^7)_1$	421.290	164-163	-616(35)
$^1S_0 - (^8, ^7)_1$	421.290	164-162	-913.2(8)
$^1S_0 - (^8, ^7)_1$	421.290	164-161	-1635.0(10)
$^1S_0 - (^8, ^7)_1$	421.290	164-160	-1895.0(20)
$^1S_0 - (^8, ^7)_1$	421.290	164-158	-2868(9)
$^1S_0 - (^8, ^7)_1$	421.290	164-156	-4300(15)
$^1S_0 - (^7, 2)_1$	456.637	164-163	-660(3)
$^1S_0 - (^7, 2)_1$	456.637	164-162	-971.0(20)
$^1S_0 - (^7, 2)_1$	456.637	164-161	-1744(3)
$^1S_0 - (^7, 2)_1$	456.637	164-160	-2020(3)
$^1S_0 - (^7, 2)_1$	456.637	164-158	-3061(4)
$^1S_0 - (^7, 2)_1$	456.637	164-156	-4604(5)
$^1S_0 - (^8, 2)_1$	564.106	164-163	-685.0(20)
$^1S_0 - (^8, 2)_1$	564.106	163-162	-328(3)
$^1S_0 - (^8, 2)_1$	564.106	162-161	-801(3)
$^1S_0 - (^8, 2)_1$	564.106	161-160	-284.0(20)
$^1S_0 - (^8, 2)_1$	564.106	164-162	-1012.0(20)
$^1S_0 - (^8, 2)_1$	564.106	162-160	-1086(5)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$

¹⁶³Dy hyperfine structure (*I* = 5/2)

level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)							
$5/5_g$	162.754(20)	1152.87(4)	F=11/2: -325.21	F=13/2: -2549.30	F=15/2: -1588.04	F=17/2: -302.63	F=19/2: +1380.44	F=21/2: +3543.30		
$7/7_g$	131.000	1901.000	F=11/2: -2270.27	F=13/2: -2005.70	F=15/2: -1450.93	F=17/2: -499.01	F=19/2: +971.23	F=21/2: +3095.25		
$9/9_g$	125.000	2901.000	F=13/2: -2129.56	F=15/2: -2087.96	F=17/2: -1654.01	F=19/2: -682.66	F=21/2: -988.20	F=23/2: +3537.75		
$5/5_g$	142.91(3)	4105.0(20)	F=13/2: -2164.17	F=15/2: -2360.07	F=17/2: -2034.75	F=19/2: -982.97	F=21/2: +1024.68	F=23/2: +4241.73		
$7/7_g$	156.50(20)	2357.0(10)	F=11/2: -2592.92	F=13/2: -2329.00	F=15/2: -1715.13	F=17/2: -618.71	F=19/2: +1110.50	F=21/2: +3640.45		
$(8, 1)_{1g}^o$	261.760	334.000	F=13/2: -6657.11	F=15/2: -4591.21	F=17/2: -2284.39	F=19/2: +216.64	F=21/2: +2923.23	F=23/2: +5804.75		
$(8, 1)_g$	89.858(17)	943.2(9)	F=9/2: -1444.76	F=11/2: -1241.27	F=13/2: -879.52	F=15/2: -298.87	F=17/2: +570.65	F=19/2: +1808.32		
$(8, 2)_g$	207.295(13)	1652.2(9)	F=13/2: -4615.44	F=15/2: -3570.97	F=17/2: -2166.94	F=19/2: -320.74	F=21/2: +2059.95	F=23/2: +5077.19		
$(8, 1)_g$	121.620(20)	1844.9(4)	F=13/2: -2407.45	F=15/2: -2065.05	F=17/2: -1431.00	F=19/2: -413.08	F=21/2: +1091.83	F=23/2: +3197.68		
$(2668)_{1g}^o$	110.3(12)	1595(57)	F=11/2: -191.53	F=13/2: -1689.04	F=15/2: -1220.66	F=17/2: -418.69	F=19/2: +818.57	F=21/2: +2164.75		

Measured A, B constants; sources in the data file